



The University of Texas at Austin
McCombs School of Business

The Road to Five Stars

FoodHub: exploratory data analysis of order, delivery and rating data

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Python Foundations • February 2024

Post Graduate Program in AI & ML: Business Applications

McCombs School of Business, The University of Texas at Austin

PROJECT

01

Python Foundations

Exploratory data analysis of 1,898 food-delivery orders: where ratings go missing, why weekday delivery lags, and what drives repeat custom.

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Executive summary — key findings

38.8%

of orders carry no rating (736 of 1,898)

0

one- or two-star ratings in the dataset

66.7%

of missing ratings sit in the 'slow' delivery bin

65%

of customers ordered once (784 of 1,200)

- Of the 10.5% of orders exceeding 60 minutes, 40.5% are unrated. Slow delivery and silence travel together.
- Weekday deliveries average 6 minutes longer than weekend (28.3 vs 22.5), on less than half the order volume.
- Weekend delivery time is fast on average but highly variable; weekday is slower and more consistent.
- The four restaurants that clear the promotion threshold are also the four most ordered.

Executive summary — recommendations

01

Two-part rating prompt

Separate scores for food and delivery, each with reason checkboxes. Minimize friction.

02

Delivery confirmation

Driver confirms handover in-app; the rating request keys off that timestamp.

03

Automated feedback request

Push, SMS or email 15–30 minutes after confirmed delivery.

04

Proactive outreach on slow orders

Contact and compensate known late deliveries before asking for a rating.

05

Monitor and respond

Track feedback and close the loop visibly so customers see it acted on.

06

Incentivize feedback

Discounts or loyalty points for completed ratings.

07

Collect delivery-chain data

Order-ready time, geography, timestamps, driver IDs — to diagnose weekday lag.

08

Loyalty and rewards program

Weekday specials and repeat-order incentives aimed at the 65% who ordered once.

09

Partner with top restaurants

Exclusive offers with the four promotion-eligible partners.

Business problem

Four issues surfaced by the data

1 · Incomplete customer feedback

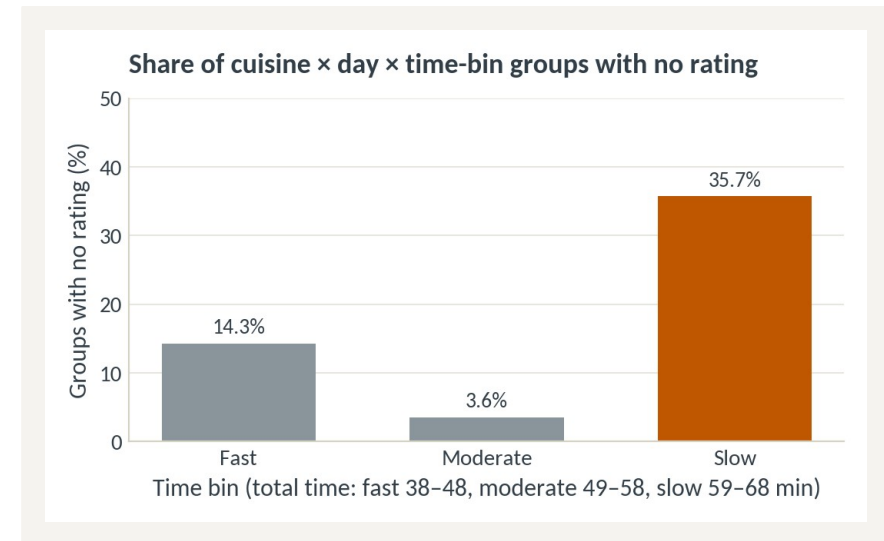
A large share of orders are never rated

- 736 of 1,898 orders (38.8%) have no rating. No order in the data received one or two stars.

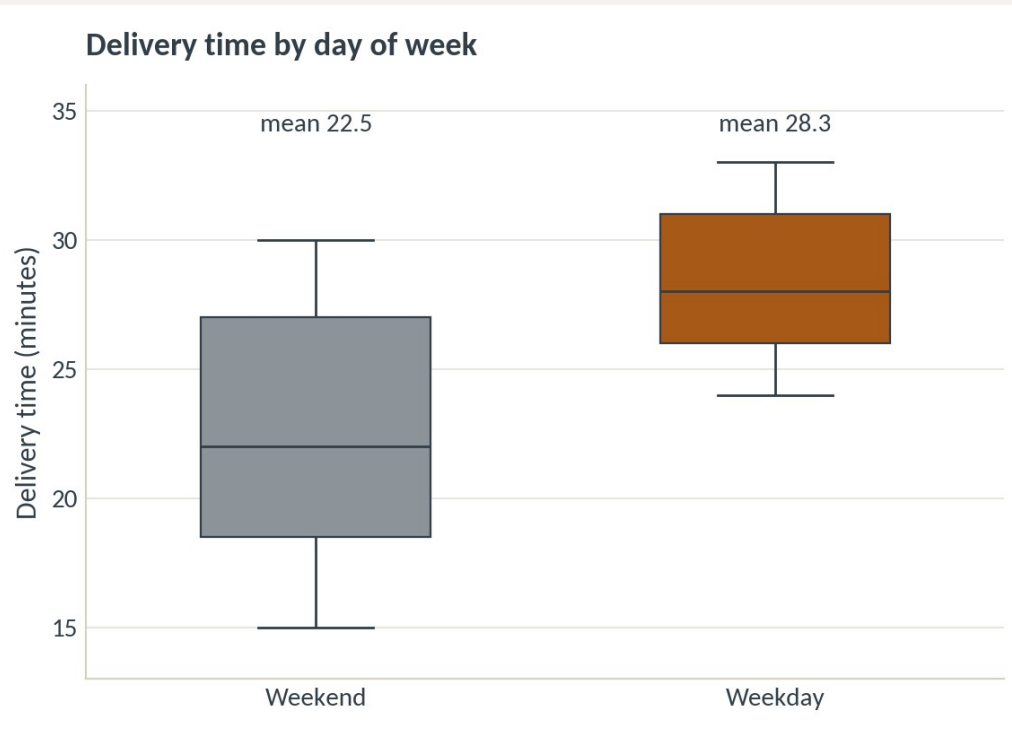


Unrated orders cluster in slow deliveries

- Dissatisfied customers may be declining to rate rather than rating low — or the rating flow may not capture low scores.



2 · Weekday delivery lag



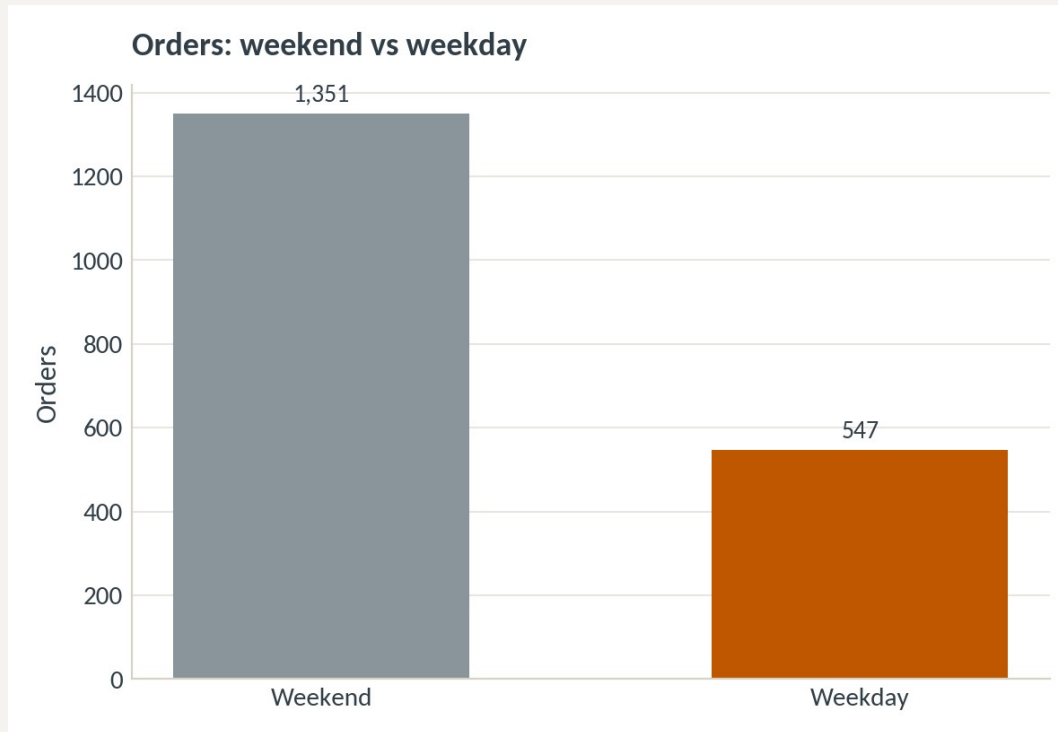
Delivery time by day of week (minutes)

Observation Weekday deliveries average 28.3 minutes against 22.5 at the weekend.

Impact Slower weekday service, inconsistent weekend service — both erode satisfaction.

Likely causes Driver availability or weekday traffic congestion. The data cannot yet distinguish them.

3 · Low weekday order volume



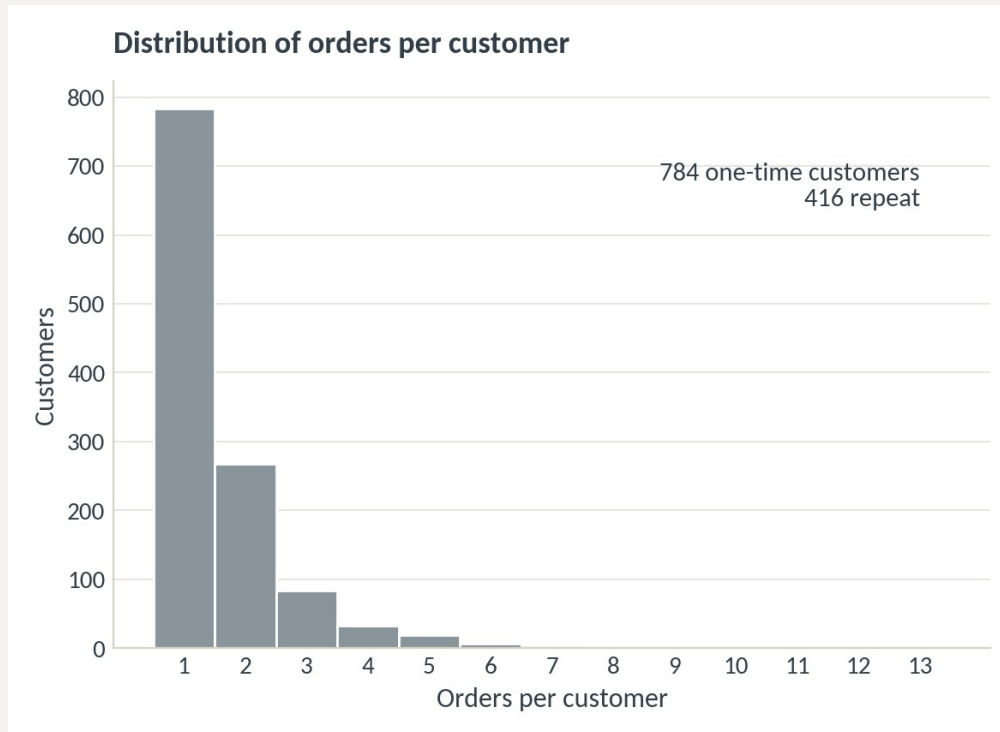
Orders: weekend vs weekday

Observation 1,351 weekend orders against 547 weekday.

Impact Revenue concentrated in two days a week.

Likely cause Customers eat in or collect on the way home rather than order delivery.

4 • Weak customer retention



Distribution of orders per customer

Observation 1,200 customers placed 1,898 orders. 784 ordered once; 416 came back, producing 698 repeat orders.

Impact Growth depends on constant acquisition; long-term planning is harder.

Likely cause Service or product issues on the first order.

Closing the feedback gap

Make rating effortless

Two-part rating prompt. One score for food, one for delivery; reason checkboxes under each.

Delivery confirmation. Driver confirms handover in-app immediately.

Automated request. Notification 15–30 minutes after confirmed delivery.

Make rating worthwhile

Proactive engagement. Reach out with compensation on known late orders before requesting feedback.

Monitor and respond. Act on feedback visibly so customers see their input matters.

Incentives. Discounts or loyalty points for completed ratings.

Diagnosing weekday delivery lag

Collect what the current dataset lacks: order-ready time, geography, timestamps and driver IDs.

Delivery-chain timing. Track driver arrival at restaurant and order-ready time for a transparent view of each stage.

Weekday pattern. Test whether traffic or driver availability explains the lag.

Proximity-ranked listings. Surface nearby, well-rated restaurants to shorten delivery.

Temporal demand. Use time and date to find peak ordering windows and plan capacity.

Driver performance. Monitor delivery times and ratings per driver; address outliers.

Retention and partner opportunities

Loyalty rewards system

- Weekday specials for loyalty members to lift the 547-order weekday baseline.
- Repeat-order incentives aimed at converting the 784 one-time customers.

Strengthen top-restaurant relationships

- Negotiate customer deals with the four promotion-eligible partners.
- Co-develop app-exclusive offers.

Dataset: 1,898 orders, 9 variables

Variables: order ID, customer ID, restaurant, cuisine, order cost, weekend/weekday, food preparation time, delivery time, rating. Derived: total time and time bins (fast, moderate, slow).

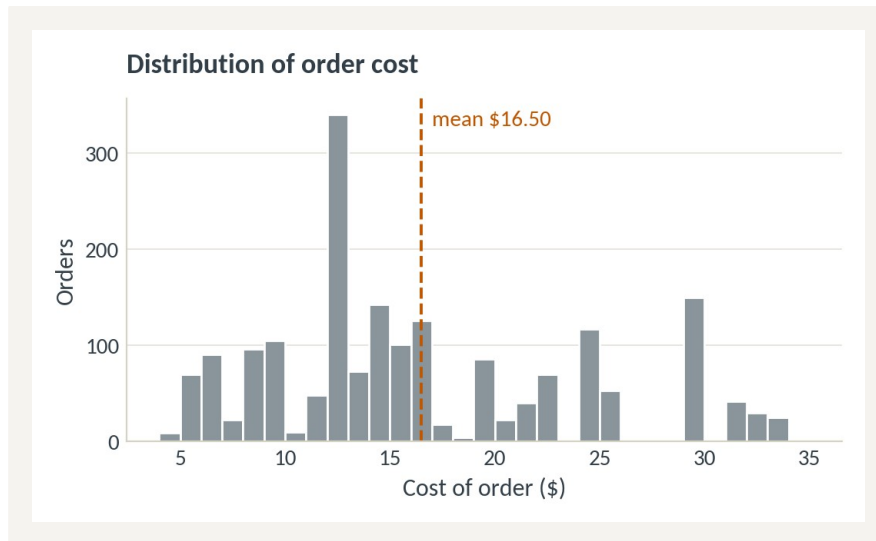
In minutes	Food prep time	Delivery time	Total time
Minimum	20	15	35
Average	27.4	24.2	51.5
Maximum	35	33	68

- Data types: float64 (1), int64 (4), object (4).
- Rating was stored as text; 736 values (38.8%) read 'Not given' and were treated as missing.

Univariate analysis — cost and cuisine

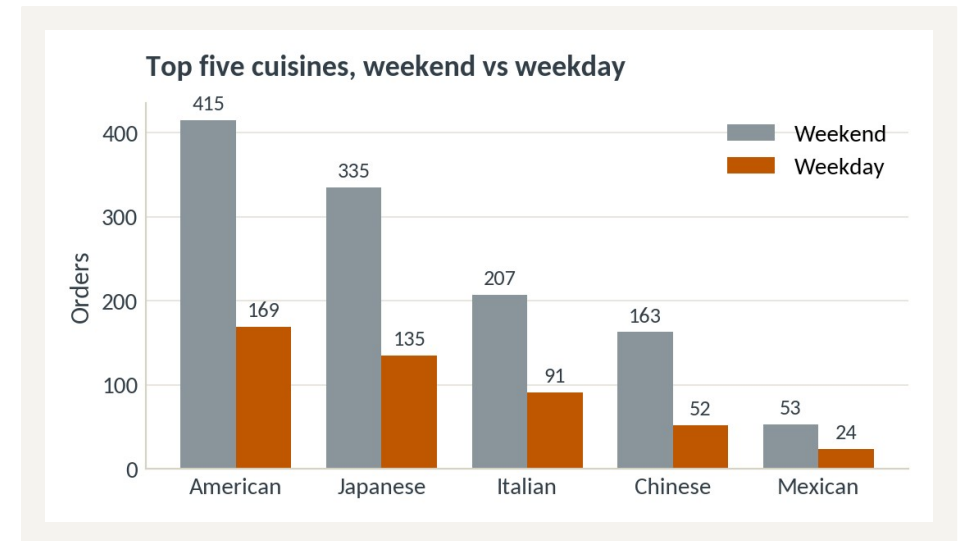
Order cost

- Average order \$16.50; \$11–12 most common. 29.2% of orders exceed \$20. The fall-off above \$12 suggests budget-constrained ordering.

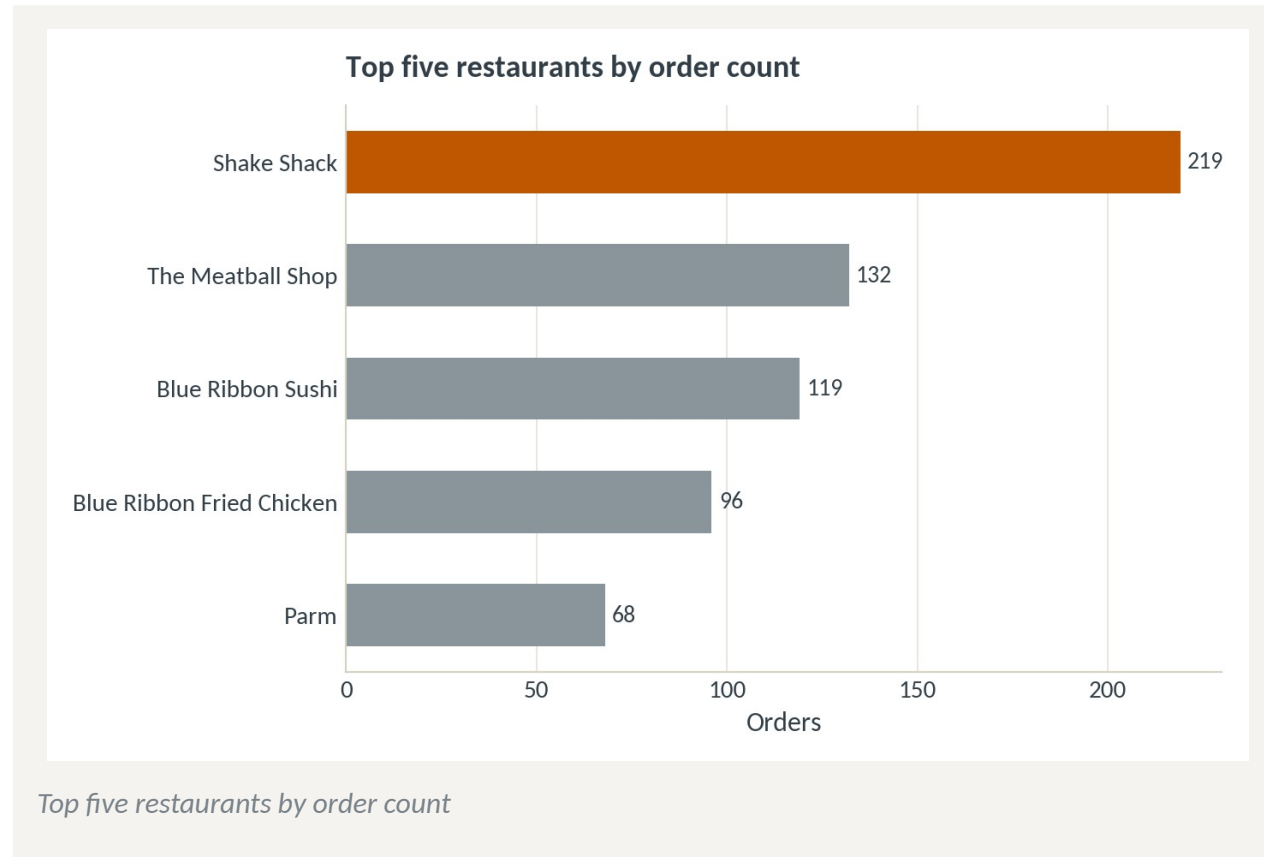


Cuisine popularity

- The top five cuisines — American, Japanese, Italian, Chinese, Mexican — hold the same rank order on weekends and weekdays.



Univariate analysis — restaurants and customers



Most-ordered restaurants. Shake Shack leads by a wide margin (219 orders); the top five together account for 634 of 1,898 orders.

Most frequent customers

- ID 52832 — 13 orders
- ID 47440 — 10 orders
- ID 83287 — 9 orders

Net revenue across all orders, at 25% commission above \$20 and 15% above \$5: \$6,166.30.

Univariate analysis — volume and timing

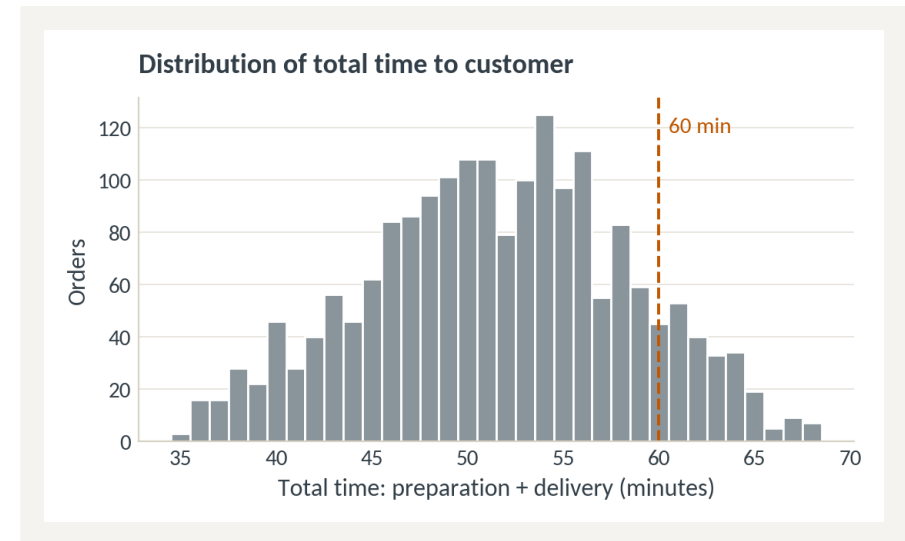
Weekday volume is low

- 1,351 weekend orders; 547 weekday. Mean delivery time is 24.2 minutes overall.

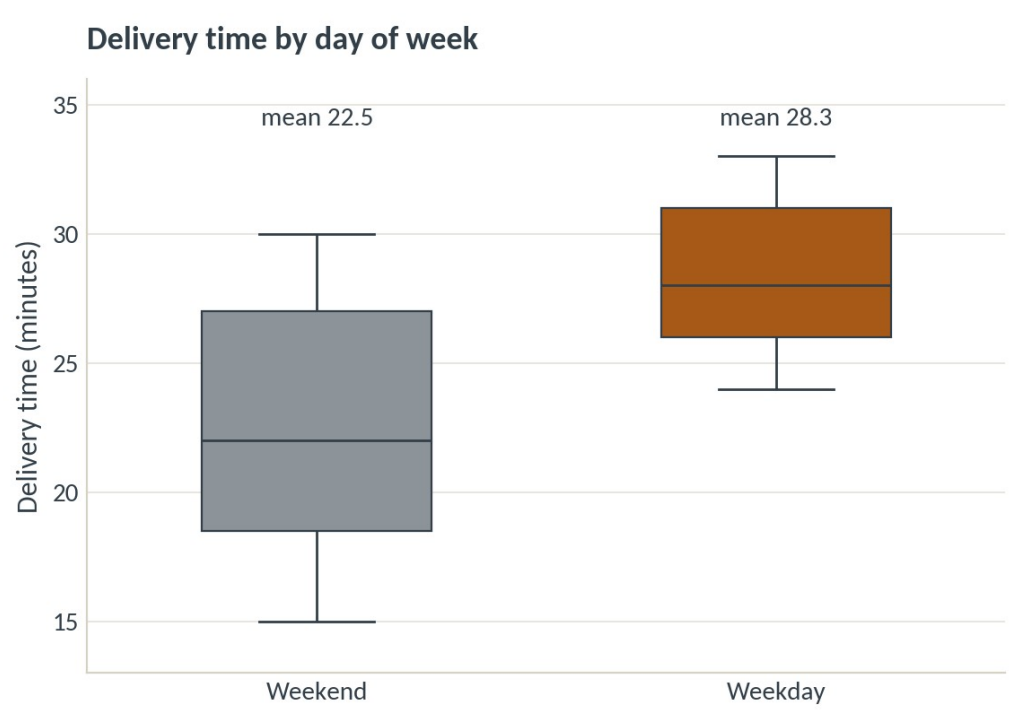


Total time to customer

- Preparation plus delivery ranges from 35 to 68 minutes, averaging 51.5. One order in ten exceeds an hour.



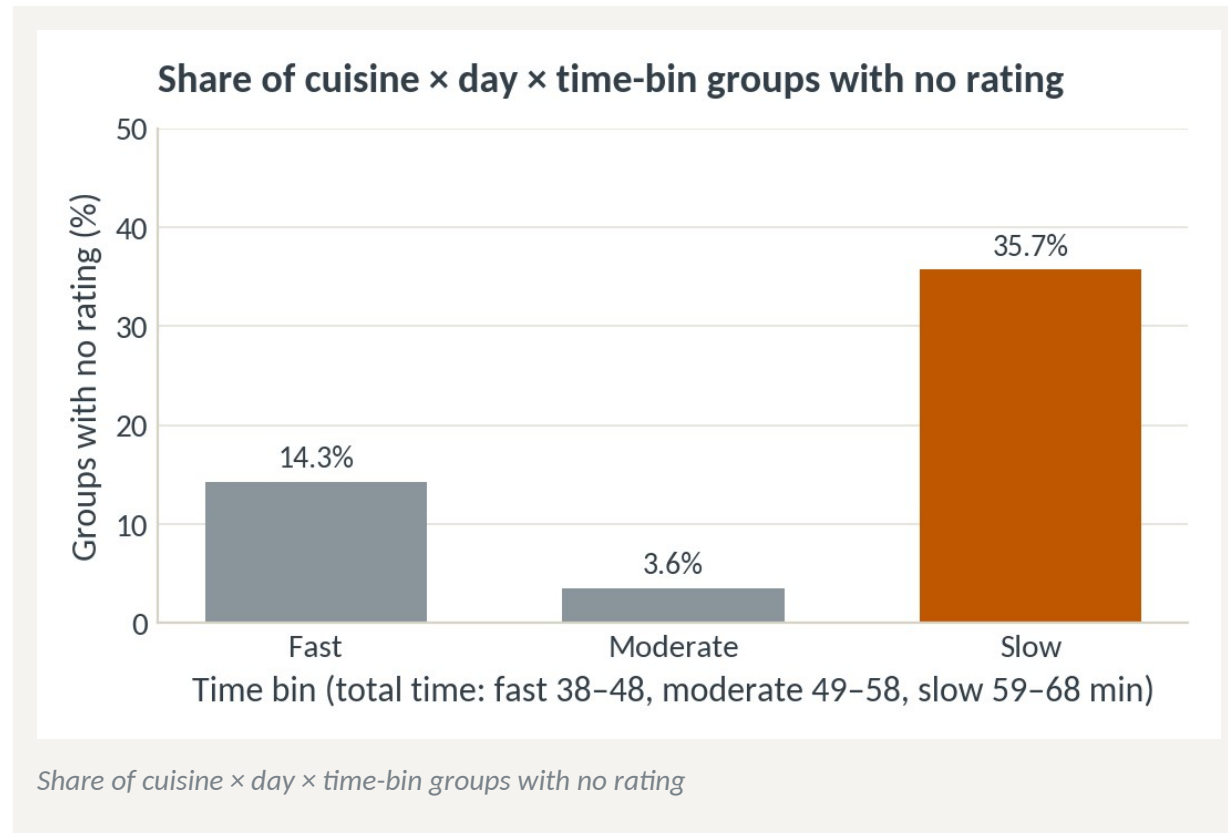
Multivariate analysis — delivery time by day



Delivery time by day of week (minutes)

- Mean delivery time differs by about 6 minutes between weekend and weekday.
- Weekend deliveries show the wider interquartile range — faster on average, less consistent.
- Weekday deliveries have the higher median but the narrower spread.
- Speed and consistency currently trade off against each other.

Multivariate analysis — where ratings go missing



66.7% of missing ratings are 'slow' orders — disproportionate against the share of slow orders overall. Total time appears to influence whether a rating is left at all.

Two candidate explanations

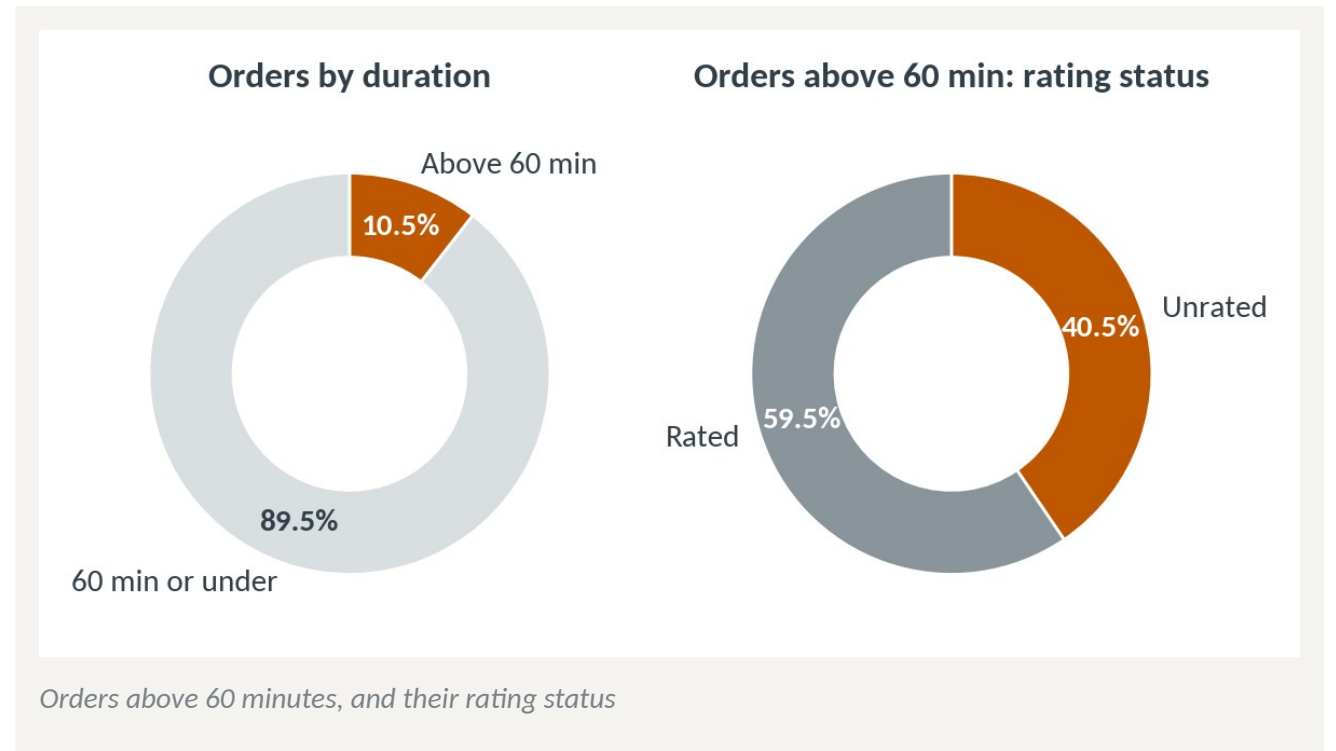
- Customers dissatisfied with slow service decline to leave a bad review.
- The survey arrives before the order does, or too long after a fast one.

Multivariate analysis — long orders and promotion eligibility

Promotion-eligible restaurants (more than 50 ratings, average above 4): The Meatball Shop (4.51), Blue Ribbon Fried Chicken (4.33), Shake Shack (4.28), Blue Ribbon Sushi (4.22). Parm, fifth by orders, falls short of 50 ratings.

10.5% of orders exceed 60 minutes; 40.5% of those are unrated. Delivery performance and rating engagement look connected.

Net revenue across all orders: \$6,166.30.



Appendix

Missing-value treatment

Missing-value treatment

With 736 ratings absent, I built a categorical structure to impute from: total time binned as fast (38–48 min), moderate (49–58) and slow (59–68); orders grouped by cuisine, day of week and time bin; each missing rating replaced by its group's mean.

Rating	Before	After imputation	Change
Not given	736	—	—
5 stars	588	667	+79
4 stars	386	1,038	+652
3 stars	188	193	+5

- The imputation loads heavily onto 4-star — 652 of 736 — because group means cluster near 4.2. It preserves row count but flattens the distribution, so the imputed column served descriptive summaries only.



Thank you

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